

A General Phase-Locked Loop to Enhance Sensorless Control of PMSM Based on Back Electromotive Force

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Abstract—To overcome the two shortcomings of the traditional phase-locked loop (PLL): incapability in realizing speed reversal and an unsatisfactory position tracking performance between dynamic and steady-state states, a new general PLL (GPLL) is proposed for permanent magnet synchronous motors sensorless control based on back electromotive force (BEMF) in both the stationary $\alpha\beta$ -axis and the rotary dq -axis. Through constructing a new input error signal, the sign of the speed variable in position error signal can be eliminated. Nevertheless, this constructed error signal contains double-fundamental frequency position information, resulting in two sets of convergence points, i.e., 0° or 180° from the actual position. To avoid incorrect convergence, a polarity correction function, based on the relationship between estimated BEMF and estimated speed, is designed to cooperate with the GPLL. Then, the sensorless drive can converge into correct position both in positive and negative rotation conditions. Furthermore, an estimated speed feedforward compensation is integrated into the GPLL, then the position tracking error can be reduced in linear acceleration and deceleration conditions, which enhances the dynamic tracking ability of the sensorless drive. Finally, the GPLL are realized both in a sliding mode observer on $\alpha\beta$ -axis and a Luenberger observer on dq -axis, and their effectiveness is verified in comparison with conventional PLL-based method and an active flux observer in terms of speed reversal ability, dynamic and steady-state performance, etc.

Index Terms—Permanent magnet synchronous motor (PMSM), phase-locked loop (PLL), polarity correction function, sensorless control, speed reversal.

I. INTRODUCTION

PERMANENT magnet synchronous motors (PMSMs) having the advantages of high power density, high efficiency, excellent dynamic and steady-state performance, etc., are widely used in household appliances, electrified transportation,

Manuscript received 21 December 2023; revised 20 April 2024 and 5 June 2024; accepted 27 June 2024. Date of publication 13 August 2024; date of current version 14 February 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 52007169, and in part by the Natural Science Foundation of Zhejiang Province under Grant LY23E070004. (Corresponding author: Chun Wu.)

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Digital Object Identifier 10.1109/TIE.2024.3423441

aerospace, and other applications [1], [2]. In all fields of PMSMs drives, it is very important to obtain rotor position/speed in real time. However, mechanical position sensors with their cable wires will increase system cost and volume and reduce system reliability [3]. Therefore, the sensorless control technology has become a research hotspot in the field of PMSMs control.

The sensorless control methods of PMSMs can be divided into two categories: 1) the high-frequency signal injection based on the inductance salience of PMSMs, which is suitable for low-speed and standstill operation [4], [5]; and 2) the observer methods based on fundamental waveform models, are widely adopted in full-speed range, especially in medium and high speeds. According to the principle of fundamental waveform models, observer methods are divided into the back electromotive force position observers (BEMF-POs) and the flux linkage position observers (F-POs), such as sliding mode observer (SMO) [6], Luenberger observer (LO) [7], and active flux observer (AFO) [8].

Among them, the BEMF-POs are widely used in the industrial fields because of its dynamic performance, antiinterference ability, reliability, and simplicity, which are more powerful than the F-POs [9]. Unfortunately, the observers have a common defect: it is very difficult to estimate the BEMF or the flux linkage accurately at very low speed due to the influence of the inverter nonlinearities and the error of the model parameters [10]. In addition, compared with the flux linkage, the BEMF is coupled with the speed information; hence, its amplitude is close to zero at very low speed or standstill, which makes it more difficult to be estimated accurately. Therefore, it is generally believed that the performance of the F-POs is better than that of the BEMF-POs under extremely low speed or speed reversal process [11]. Furthermore, since the BEMF is coupled with the speed, it will be affected by the speed symbol sign. The BEMF-POs have to tackle the speed symbol contained in the BEMF to achieve speed reversal successfully. However, it is not necessary for the AFO methods because the flux linkage is independent of speed symbol [12]. Therefore, many scholars hope to improve the algorithm to eliminate the speed symbol in the BEMF estimation; however, even the latest improvements have some defects and not solved them thoroughly. This article is dedicated to address the defects and design a new phase-locked loop (PLL) to realize better low-speed performance and more reliable speed reversal based on the conventional BEMF-POs.

It is well known that BEMF-POs can be implemented either on the stationary $\alpha\beta$ -axis or the rotating dq -axis, which depends on the selection of the mathematic model of PMSMs. At first, the $\arctan$ function was widely used to obtain the rotor position

directly from the BEMF estimated by BEMF-POs on $\alpha\beta(dq)$ -axis [13]. However, the $\arctan$ function will cause noise and harmonics, which will affect the accuracy of position tracking, especially when the motor speed crosses zero [12]. To solve this problem, the PLL replaced $\arctan$ method because of its filter characteristic and moderate tracking performance. However, the traditional PLL for BEMF-POs has two shortcomings: 1) incapability in realizing speed reversal [12]; and 2) an unsatisfactory position tracking performance both in dynamic and steady states [14]. To solve the above problems, scholars have proposed several modifications and improvements for the PLL.

With respect to the first defect, an intuitive method is to multiply the sign of the estimated speed with the estimated BEMF or position error. This method is referred to as the speed-symbol-PLL (S-PLL) in this article. However, this method has big drawbacks: when the motor speed is extremely low since the speed estimated by the BEMF-POs is affected by steady-state errors, the speed symbol will switch between positive value and negative value with high frequency. If a wrong speed symbol is used for the PLL in several consecutive control cycles, the PLL's input will always be the opposite of the true BEMF, resulting in a "positive-feedback system," i.e., the greater input, the larger position tracking error, leading to lock the rotor and even get out of control [10].

To completely solve the effect of the speed symbol, some experts proposed an algorithm to eliminate the speed symbol in the BEMF based on $\arctan$ function, but it was easy to be affected by system noise due to the use of $\arctan$ function [15]. A novel PLL based on $\sin$ and $\cos$ functions was proposed, which eliminated the speed symbol in the position error signal and could solve the problem of speed reversal and avoid the use of $\arctan$ function [16]. However, the authors [15], [16] have not reduced the dynamic tracking error. A fractional order PLL with a fractional order proportional-integral (PI) was presented, and the position tracking error in the dynamic processes could be reduced effectively [17], but it has not changed the essence of the traditional PLL as a second-order system, so the position tracking error in dynamic processes cannot be eliminated. [14] constructed a high-order PLL by adding an estimated speed feedforward compensation, which improved the position tracking error in dynamic process while the steady-state error was almost unchanged. However, the authors [14], [17] have not concerned and solved the speed reversal problem. Furthermore, a dual PLL (DPLL) was proposed in [12] to eliminate the speed symbol in the position error signal by $\sin$ and $\cos$ functions, which could realize speed reversal, the DPLL also eliminates the dynamic tracking error in the linear acceleration or deceleration of speed by constructing high-order PLL. However, the error signals of the improved PLL proposed in [12], [15], and [16] all contain double-fundamental frequency position information, so they inevitably have two different convergence points, i.e., 0° and 180° , and only 0° is the correct convergence point for the PLLs. Li et al. [18] analyzed the influence of incorrect convergence on motor control performance; however, it did not give a solution.

All of the above literatures have made significant efforts to solve the existing shortcomings in the traditional PLL based on the BEMF. Yet even the latest improvements have flaws. To solve the two shortcomings simultaneously and thoroughly, a

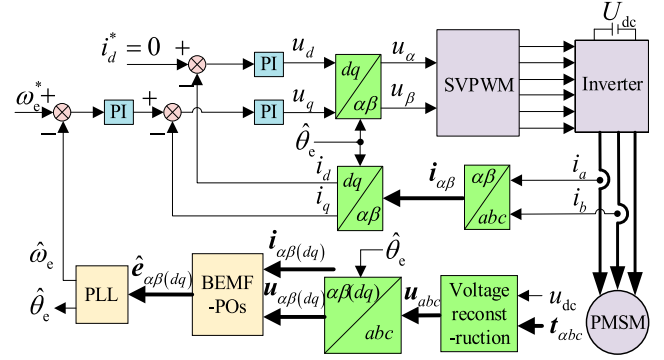


Fig. 1. PMSM sensorless control structure block diagram.

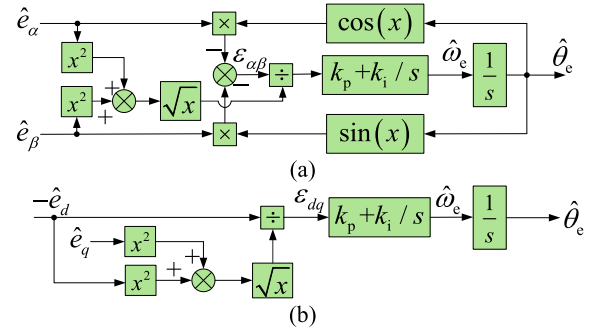


Fig. 2. Block diagrams of the traditional PLL on (a) $\alpha\beta$ - and (b) dq -axes.

new general PLL (GPLL) is proposed in both the stationary $\alpha\beta$ -axis and the rotating dq -axis. First, the speed symbol in the position error signal is eliminated by constructing a new input error signal. Considering that the new error signal also contains double-fundamental frequency position information, a polarity correction function is designed based on the relationship between the estimated BEMF and the estimated speed, which avoids incorrect convergence and can achieve accurate position tracking under both positive and negative rotation conditions. Furthermore, the polarity correction function also avoids the existence of the speed symbol. Second, adding the estimated speed feedforward compensation to the GPLL to further reduce the position tracking error in dynamic process. Finally, two typical BEMF-POs (i.e., the SMO on $\alpha\beta$ -axis and the LO on dq -axis) based on the GPLL are compared with the BEMF-POs based on the S-PLL and the AFO method under various conditions, and the feasibility and effectiveness of the proposed GPLL are further verified by comprehensive experiments.

II. PERFORMANCE ANALYSIS OF BEMF-POs BASED ON THE TRADITIONAL PLL

Fig. 1 shows the block diagram of the sensorless control of PMSMs based on BEMF-POs with the traditional PLL, and the two shortcomings of the traditional PLL are as follows.

A. Ability of Speed Reversal

Fig. 2(a) and 2(b) shows the block diagrams of the traditional PLL, which is implemented on $\alpha\beta$ - and dq -axes, respectively.

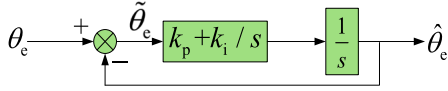


Fig. 3. Equivalent diagram of the traditional PLL.

$\hat{e}_{\alpha\beta(dq)}$ is the estimated BEMF on $\alpha\beta(dq)$ -axis; “” represents the estimated variables; $\hat{\theta}_e$ is the estimated position; $e_{\alpha\beta(dq)}$ is the position error signal; k_p and k_i are the PI parameters; and $\hat{\omega}_e$ is the estimated angular speed. Since

$$\begin{cases} \hat{e}_\alpha = -K_e \sin \theta_e \\ \hat{e}_\beta = K_e \cos \theta_e \end{cases} \quad (1)$$

where θ_e is the actual rotor position; $K_e = \omega_e \Psi_f$, ω_e is the actual speed; and Ψ_f is the actual permeant magnet flux linkage.

The error signal of the traditional PLL on $\alpha\beta$ -axis is as follows:

$$\begin{aligned} e_{\alpha\beta} &= \frac{-\hat{e}_\alpha \cos \hat{\theta}_e - \hat{e}_\beta \sin \hat{\theta}_e}{\sqrt{\hat{e}_\alpha^2 + \hat{e}_\beta^2}} \\ &= \frac{K_e (\sin \theta_e \cos \hat{\theta}_e - \cos \theta_e \sin \hat{\theta}_e)}{\sqrt{\hat{e}_\alpha^2 + \hat{e}_\beta^2}} \approx \text{sign}(\omega_e) \tilde{\theta}_e \end{aligned} \quad (2)$$

where $\tilde{\theta}_e = \theta_e - \hat{\theta}_e$ is the position tracking error.

According to (1), since $\hat{e}_{\alpha\beta}$ is affected by the sign of ω_e , if the motor spins in positive direction, i.e., $\omega_e > 0$, then, $e_{\alpha\beta} \approx \tilde{\theta}_e$, so $(\tilde{\theta}_e, \tilde{\omega}_e)$ of the traditional PLL will converge to the points $(2k\pi, 0)$, where $k = 0, \pm 1, \pm 2 \dots$, the rotor position can be correctly tracked ($\tilde{\omega}_e$ is the speed estimation error); When the motor is reversed, i.e., $\omega_e < 0$, then, $e_{\alpha\beta} \approx -\tilde{\theta}_e$, so $(\tilde{\theta}_e, \tilde{\omega}_e)$ will converge to $((2k+1)\pi, 0)$, which has 180° steady-state error from the actual position, and induces losing control.

The error signal of the traditional PLL on dq -axis is as follows:

$$e_{dq} = \frac{-\hat{e}_d}{\sqrt{\hat{e}_d^2 + \hat{e}_q^2}} = \frac{K_e}{\sqrt{\hat{e}_d^2 + \hat{e}_q^2}} \sin \tilde{\theta}_e \approx \text{sign}(\omega_e) \tilde{\theta}_e. \quad (3)$$

Similarly, the traditional PLL on dq -axis can only track the rotor position when the motor runs in a positive direction.

To solve this problem, scholars generally used to multiply the speed symbol $\text{sign}(\hat{\omega}_e)$, and the $e_{\alpha\beta(dq)}$ is obtained

$$e_{\alpha\beta(dq)} = \text{sign}(\hat{\omega}_e) \cdot \text{sign}(\omega_e) \tilde{\theta}_e. \quad (4)$$

This is the S-PLL mentioned in the introduction. However, we can see from (4) that the S-PLL is only feasible when the sign of $\hat{\omega}_e$ is exactly the same as the sign of ω_e . When the motor is running at an extremely low speed or is crossing-zero speed, the sign of $\hat{\omega}_e$ is likely to be opposite to the sign of ω_e sometimes because of the steady-state errors. At this time, $e_{\alpha\beta(dq)} \approx -\tilde{\theta}_e$, so $(\tilde{\theta}_e, \tilde{\omega}_e)$ will converge to the points $((2k+1)\pi, 0)$, causing the S-PLL to form a “positive-feedback system,” and resulting in divergence of the system. Therefore, the S-PLL faces a great risk under extremely low speed and speed reversal conditions due to the influence of the speed symbol.

B. Dynamic and Steady-State Performance Analysis

The equivalent diagram of the traditional PLL based on either $\alpha\beta$ - or dq -axis is demonstrated in Fig. 3, and the open-loop transfer function of the traditional PLL is expressed as

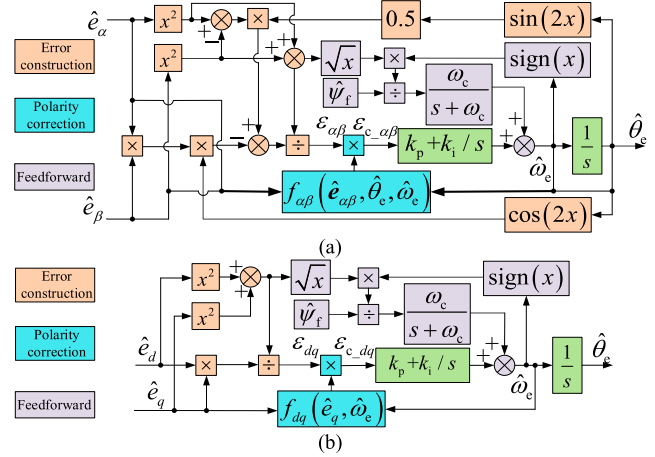


Fig. 4. Block diagrams of the proposed GPLL. (a) $\alpha\beta$. (b) dq .

$$G_O(s) = \frac{\hat{\theta}_e(s)}{\theta_e(s)} = \frac{k_p s + k_i}{s^2}. \quad (5)$$

According to (5), the traditional PLL is a typical second-order system. Therefore, when the motor is in dynamic processes, zero dynamic position tracking error cannot be realized due to its limited tracking bandwidth of the PI controllers. This analysis will be presented as follows.

The error transfer function of the traditional PLL is shown as

$$G_{\text{err}}(s) = \frac{\theta_e(s) - \hat{\theta}_e(s)}{\theta_e(s)} = \frac{s^2}{s^2 + k_p s + k_i}. \quad (6)$$

According to the final-value theorem, if the speed is a ramp signal, i.e. $\omega_e = K_a t$, and $\theta_e = 0.5 K_a t^2$, the steady-state error

$$e_{\text{ss}} = \lim_{s \rightarrow 0} s G_{\text{err}}(s) \frac{K_a}{s^3} = \frac{K_a}{k_i}. \quad (7)$$

From the above equation, during linear acceleration and deceleration processes, the dynamic position tracking error is proportional to the acceleration coefficient K_a . Although increasing k_i can reduce this dynamic error, it will lower the phase margin and degrade the steady-state performance. Hence, how to completely achieve speed reversal and zero dynamic position tracking error is still an unsolved problem.

III. THE PROPOSED GPLL

To solve the above two shortcomings, a novel GPLL is proposed, which can be implemented both $\alpha\beta$ - and dq -axes. Its structure is drawn in Fig. 4(a) and 4(b), which are based on $\alpha\beta$ -axis and dq -axis, respectively.

A. Error Signal Construction of the GPLL

To eliminate the sign of speed variable and avoid the influence of $\tan$ function, the input error signal of the GPLL on $\alpha\beta$ -axis adopted the reconstruction method proposed in [12]

$$\begin{aligned}
\varepsilon_{\alpha\beta} &= \frac{-\hat{e}_\alpha \hat{e}_\beta \cos(2\hat{\theta}_e) + \frac{(\hat{e}_\alpha^2 - \hat{e}_\beta^2)}{2} \sin(2\hat{\theta}_e)}{(\hat{e}_\alpha^2 + \hat{e}_\beta^2)} \\
&= \frac{K_e^2 \left(\sin(2\hat{\theta}_e) \cos(2\hat{\theta}_e) - \cos(2\hat{\theta}_e) \sin(2\hat{\theta}_e) \right)}{2(\hat{e}_\alpha^2 + \hat{e}_\beta^2)} \\
&\approx \frac{1}{2} \sin(2\tilde{\theta}_e) \approx \tilde{\theta}_e.
\end{aligned} \quad (8)$$

According to (8), the error signal of the GPLL in dq -axis is derived as

$$\begin{aligned}
\varepsilon_{dq} &= \frac{\hat{e}_d \times \hat{e}_q}{(\hat{e}_d^2 + \hat{e}_q^2)} = \frac{K_e^2 \left(\sin(\tilde{\theta}_e) \times \cos(\tilde{\theta}_e) \right)}{(\hat{e}_d^2 + \hat{e}_q^2)} \\
&\approx \frac{1}{2} \sin(2\tilde{\theta}_e) \approx \tilde{\theta}_e.
\end{aligned} \quad (9)$$

It is observed that the input error signals of the GPLL are no longer affected by the speed symbol. However, double-fundamental frequency position information ($2\tilde{\theta}_e$) is brought into the error signals, so it is necessary to analyze the convergence points of the GPLL through the Lyapunov stability theory [18].

First, from the structure of the GPLL, it is obtained that

$$\begin{cases} \dot{\hat{\theta}}_e = \int \hat{\omega}_e dt \\ \dot{\hat{\omega}}_e = k_p \varepsilon + k_i \int \varepsilon dt \end{cases} \quad (10)$$

Since the K_e^2 is always greater than 0, $\varepsilon = \sin(2\tilde{\theta}_e)/2$, which is substituted into (10). Then, a large signal model of the GPLL is obtained (assume $\dot{\omega}_e = 0$ at steady-state)

$$\begin{cases} \dot{\tilde{\theta}}_e = \tilde{\omega}_e \\ \dot{\tilde{\omega}}_e = -k_p \cos(2\tilde{\theta}_e) \tilde{\omega}_e - \frac{k_i}{2} \sin(2\tilde{\theta}_e) \end{cases} \quad (11)$$

According to the definition of the equilibrium point of the nonlinear system, the equilibrium point $(\tilde{\theta}_e^*, \tilde{\omega}_e^*)$ of the nonlinear system (11) should meet the condition that the derivative is zero, that is,

$$\begin{cases} 0 = \tilde{\omega}_e^* \\ 0 = -k_p \cos(2\tilde{\theta}_e^*) \tilde{\omega}_e^* - \frac{k_i}{2} \sin(2\tilde{\theta}_e^*) \end{cases} \quad (12)$$

Therefore, system (11) has an infinite equilibrium points, i.e., $(n\pi/2, 0)$, $n = 0, \pm 1, \pm 2, \dots$. Define $\tilde{\theta}_n = \tilde{\theta}_e - n\pi/2$, $\tilde{\omega}_n = \tilde{\omega}_e - 0$, and the system (11) is approximated in a first-order linear approximation at the equilibrium point, expressed as

$$\begin{pmatrix} \dot{\tilde{\theta}}_n \\ \dot{\tilde{\omega}}_n \end{pmatrix} = J(\tilde{\theta}_n, \tilde{\omega}_n) \begin{pmatrix} \tilde{\theta}_n \\ \tilde{\omega}_n \end{pmatrix} \quad (13)$$

where $J(\tilde{\theta}_n, \tilde{\omega}_n)$ is the Jacobian matrix of the system, and

$$J(\tilde{\theta}_n, \tilde{\omega}_n) = \begin{bmatrix} 0 & 1 \\ -k_i \cos(n\pi) & -k_p \cos(n\pi) \end{bmatrix}. \quad (14)$$

When $n = 2k + 1$, $k = 0, \pm 1, \pm 2, \dots$, i.e., $(\tilde{\theta}_e = k\pi + \pi/2)$, $J(\tilde{\theta}_e, \tilde{\omega}_e)$ has one eigenvalue with positive real part and another

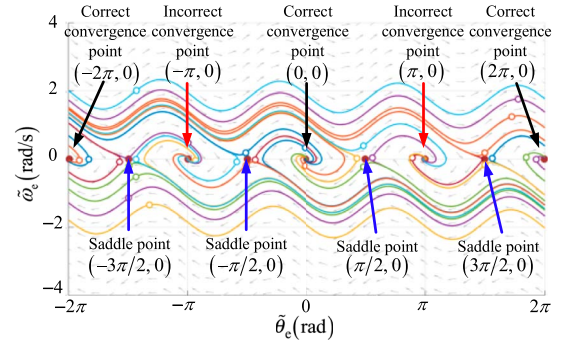


Fig. 5. Equilibrium phase diagram of the GPLL.

eigenvalue with negative real part. Therefore, regardless of the rotation direction, the equilibrium points $(k\pi + \pi/2, 0)$ are saddle points [18], which means that the system is unstable and cannot converge, i.e., incorrect position tracking.

Similarly, when $n = 2k$, $k = 0, \pm 1, \pm 2, \dots$, i.e., $(\tilde{\theta}_e = k\pi)$, $J(\tilde{\theta}_e, \tilde{\omega}_e)$ has two eigenvalues with negative real parts. Therefore, regardless of the rotation direction, the equilibrium points $(k\pi, 0)$ are convergence points [18], which means that the system is stable and can converge. Furthermore, this set of convergence points can be divided into two classes: $(2k\pi, 0)$, $k = 0, \pm 1, \pm 2, \dots$ is the correct convergence point, which means that the rotor position can be correctly tracked; and $((2k + 1)\pi, 0)$, $k = 0, \pm 1, \pm 2, \dots$ is the incorrect convergence point, which means that there is a steady-state error of 180° between the tracking rotor position and the actual position, and will lead to motor control failure.

In summary, the equilibrium points of the system can be divided into three groups: group I $(k\pi + \pi/2, 0)$, $k = 0, \pm 1, \pm 2, \dots$ is the saddle points; group II $(2k\pi, 0)$ is the correct convergence point; and Group III $((2k + 1)\pi, 0)$ is the incorrect convergence point. To analyze the relationship between saddle point and convergence point intuitively, the equilibrium point phase diagram of the GPLL is shown in Fig. 5. It can be seen from the track of the arrow in the background that when $n = 2k$, the phase trajectory near the equilibrium points $(\tilde{\theta}_e, \tilde{\omega}_e) = (k\pi, 0)$ gathers helically toward these points, so these points are convergence points. When $n = 2k + 1$, the phase trajectory near the equilibrium point $(\tilde{\theta}_e, \tilde{\omega}_e) = (k\pi + \pi/2, 0)$ is moving away from these points and toward the convergence points nearby, so these points are saddle points. In addition, it can also be seen from [18] that the points near the saddle points will automatically move to the convergence points.

Therefore, a PLL containing $2\tilde{\theta}_e$ in the error signal will only converge in two ways: one will converge to the correct convergence point of Group II, which is the same as the actual position, and the other will converge to the incorrect convergence point of Group III, which has a steady-state error of 180° from the actual position. Until now, although some literatures have tried to change the error signal, which includes the $2\tilde{\theta}_e$ information due to the newly constructed PLL's error [12], [15], [16], however, it has not been well addressed to avoid the incorrect convergence due to the $2\tilde{\theta}_e$ information. To work out this problem, a polarity correction function integrated into the GPLL is proposed in the following.

B. Polarity Correction to Achieve Speed Reversal

From the above analyses of the GPLL, if $(\tilde{\theta}_e, \tilde{\omega}_e)$ converges to $(2k\pi, 0)$, we have $\cos(\tilde{\theta}_e) = 1$; otherwise, $(\tilde{\theta}_e, \tilde{\omega}_e)$ converges to $((2k+1)\pi, 0)$, it means $\cos(\tilde{\theta}_e) = -1$. Based on this principle, we can judge the correctness of the convergence points $(\tilde{\theta}_e, \tilde{\omega}_e)$ by the sign of the $\cos(\tilde{\theta}_e)$.

Therefore, for the GPLL on $\alpha\beta$ -axis, the $\cos(\tilde{\theta}_e)$ can be calculated by

$$f_{\cos_{\alpha\beta}}(\hat{e}_{\alpha\beta}, \hat{\theta}_e, \hat{\omega}_e) = (-\hat{e}_x \sin(\hat{\theta}_e) + \hat{e}_y \cos(\hat{\theta}_e)) \times \hat{\omega}_e = \hat{\omega}_e^2 \psi_f \cos(\tilde{\theta}_e). \quad (15)$$

Similarly, for the GPLL on dq -axis, the $\cos(\tilde{\theta}_e)$ can be calculated by

$$f_{\cos_{dq}}(\hat{e}_q, \hat{\omega}_e) = \hat{e}_q \times \hat{\omega}_e = \hat{\omega}_e^2 \psi_f \cos(\tilde{\theta}_e). \quad (16)$$

Therefore, if the sign of $f_{\cos_{\alpha\beta}(dq)}$ is negative, it can be derived that $(\tilde{\theta}_e, \tilde{\omega}_e)$ converges to $((2k+1)\pi, 0)$, which are incorrect convergence points. Hence, we design a polarity correction function $f_{\alpha\beta(dq)}$ to act on the error signal $\varepsilon_{\alpha\beta(dq)}$ so that $(\tilde{\theta}_e, \tilde{\omega}_e)$ can quickly return to the correct convergence domain of $(2k\pi, 0)$, shown as

$$\varepsilon_{c_{\alpha\beta(dq)}} = f_{\alpha\beta(dq)} \times \varepsilon_{\alpha\beta(dq)} \quad (17)$$

where $\varepsilon_{c_{\alpha\beta(dq)}}$ is the reconstructed error signal after introducing the polarity correction function on $\alpha\beta(dq)$ -axis. The designed polarity correction functions $f_{\alpha\beta}(\hat{e}_{\alpha\beta}, \hat{\theta}_e, \hat{\omega}_e)$ on $\alpha\beta$ -axis and the $f_{dq}(\hat{e}_q, \hat{\omega}_e)$ on dq -axis are shown as follows:

$$f_{\alpha\beta}(\hat{e}_{\alpha\beta}, \hat{\theta}_e, \hat{\omega}_e) = P_{\text{cor}}(\text{sign}(f_{\cos_{\alpha\beta}}(\hat{e}_{\alpha\beta}, \hat{\theta}_e, \hat{\omega}_e)) - 1) + 1 \quad (18)$$

$$f_{dq}(\hat{e}_q, \hat{\omega}_e) = P_{\text{cor}}(\text{sign}(f_{\cos_{dq}}(\hat{e}_q, \hat{\omega}_e)) - 1) + 1 \quad (19)$$

where P_{cor} is the correction gain and should be met the requirement $P_{\text{cor}} > 0.5$. From (18) and (19), if the $\text{sign}(f_{\cos_{\alpha\beta(dq)}})$ is positive, we have $f_{\alpha\beta(dq)} = 1$. Otherwise, we have $f_{\alpha\beta(dq)} = 1 - 2P_{\text{cor}}$, which means larger P_{cor} causes a faster return from the incorrect convergence points to correct convergence points. Therefore, if the system converges to correct convergence points, then $\text{sign}(f_{\cos_{\alpha\beta(dq)}}) = 1$, and the $f_{\alpha\beta(dq)}$ goes back to 1 again. Therefore, the polarity correction function only acts during the transient process from the incorrect convergence domain to the correct convergence domain and does not affect the tracking performance at all in the correct convergence domain because of $f_{\alpha\beta} = 1$.

In addition, it can also be seen from (15) that the proposed polarity correction function is completely unaffected by the speed symbol, that is, it solves the problem of two sets of convergence points and not decreases the magnitude of the reconstructed error signal itself. Furthermore, it can make the GPLL very robust in determining the convergence domain by judging the sign of $\cos(\tilde{\theta}_e)$, unlike S-PLL using the estimated speed sign that is not always accurate in low-speed processes.

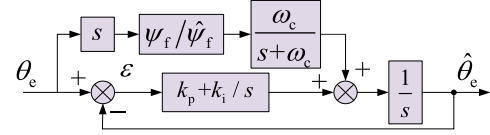


Fig. 6. Equivalent diagram of the GPLL.

To sum up, the polarity correction function is introduced to ensure that $(\tilde{\theta}_e, \tilde{\omega}_e)$ converges to the correct points $(2k\pi, 0)$ no matter what the initial conditions are.

C. Estimated Speed Feedforward Compensation to Enhance Dynamic Performance

To solve the second problem of the traditional PLL that the dynamic position tracking error is also unsatisfactory, we add an estimated speed feedforward compensation to the GPLL on $\alpha\beta(dq)$ -axis. Since the feed forward compensation also uses the estimated speed, it also needs to be multiplied by the sign of the estimated speed, shown in Fig. 4.

Whether on $\alpha\beta$ - or dq -axis, the equivalent diagram of the GPLL with the feedforward compensation is shown in Fig. 6, where $\hat{\psi}_f$ is the flux linkage used by the algorithm and ω_c is the cutoff frequency of the low-pass filter (LPF).

The open-loop transfer function of the GPLL is derived as

$$G_{O-NEW}(s) = \frac{(k_p + \omega_c \psi_f / \hat{\psi}_f) s^2 + (k_i + k_p \omega_c) s + k_i \omega_c}{s^2 (s + \omega_c)}. \quad (20)$$

It is observed that the GPLL is a third-order system. To analyze the steady-state performance of the GPLL, its error transfer function is obtained as follows:

$$G_{err-NEW}(s) = \frac{s^3 + \omega_c (1 - \psi_f / \hat{\psi}_f) s^2}{(s + \omega_c) (s^2 + k_p s + k_i)}. \quad (21)$$

Ideally, assuming $\Psi_f / \hat{\Psi}_f = 1$ and the motor is in a linear acceleration or deceleration state, the dynamic position tracking error can be calculated by the final-value theorem as

$$e_{ss-NEW} = \lim_{s \rightarrow 0} s G_{err-NEW}(s) \frac{K_a}{s^3} = 0. \quad (22)$$

According to (22), the GPLL with the feedforward compensation can achieve zero dynamic position tracking error during a linear acceleration or deceleration, which cannot be achieved in the traditional PLL.

In summary, the proposed GPLL is a general PLL frame and is applicable to all BEMF-POs both on $\alpha\beta$ - and dq -axes. It can not only solve the problem of the speed reversal of the motor but also reduce the dynamic error of the position tracking in transients while maintaining steady-state tracking performance.

D. Design of GPLL Parameters and Analyze Their Effects on Dynamic and Steady-State Performance

1) *Feedforward Compensation Parameters Design of the GPLL:* It can be seen from (20) that when the polarity correction function does not work, the position tracking performance of the

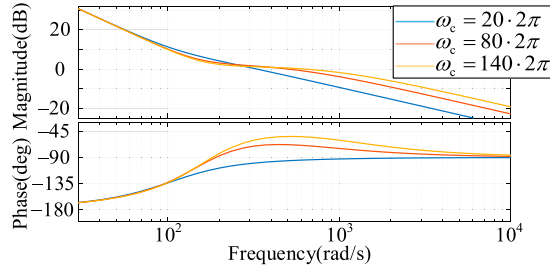


Fig. 7. Bode diagrams of the GPLL with different ω_c .

GPLL is related to its PI parameters and the ω_c in the feedforward path. In the process of designing the GPLL, the basic principle is to ensure that the steady-state performance of the traditional PLL and the GPLL are basically the same. So, the cross-over frequency of the GPLL ω_{c_PLL} in (20) is chosen to be equal to the cross-over frequency of the traditional PLL ω_{PLL} in (5) (in engineering, the cross-over frequency of the open-loop transfer function can be treated as approximately equivalent to the bandwidth of the PLL). We get the cross-over frequency of the GPLL ω_{c_PLL} through (assumed $\Psi_f/\hat{\Psi}_f = 1$)

$$|G_{O-NEW}(j\omega_{c_PLL})| = 1. \quad (23)$$

According to [7], the bandwidth of the PLL needs to be four to ten times larger than that of speed controller ω_{spd} . In this article, $\omega_{spd} = 20 \times 2\pi$ rad/s, so, let $\omega_{PLL} = 4 \times \omega_{spd} = 80 \times 2\pi$ rad/s. Then, k_p and k_i can be obtained by substituting ω_{PLL} into (5)

$$|G_O(j\omega_{PLL})| = \left| \frac{-j\omega_{PLL}k_p + k_i}{-\omega_{PLL}^2} \right| = 1. \quad (24)$$

It follows that $k_p = \sqrt{3}\omega_{PLL}/2$ and $k_i = \omega_{PLL}^2/2$.

To ensure that the steady-state performance of the GPLL is not affected, similarly set $\omega_{c_PLL} = 4 \times \omega_{spd} = 80 \times 2\pi$ rad/s. In addition, due to the addition of the feedforward, k_p and k_i of the GPLL can be smaller than those of the traditional PLL, that is, let $k_p = \sqrt{3} \times (40 \times 2\pi)/2$ and $k_i = (40 \times 2\pi)^2/2$. Finally, by substituting ω_{c_PLL} , k_p and k_i into (23), relatively, we can obtain $\omega_c = 80 \times 2\pi$ rad/s. This set of parameters can not only improve the dynamic performance but also ensure that the steady-state performance is not affected.

Fig. 7 shows the Bode diagram of the GPLL under different ω_c at the conditions of $\omega_{PLL} = 40 \times 2\pi$ rad/s and $\Psi_f/\hat{\Psi}_f = 1$. From Fig. 7, with the increase of ω_c , the bandwidth of the system increases. The larger the bandwidth, the better the dynamic responds performance of the system, but the steady-state performance will decrease due to the decrease of amplitude margin and phase margin.

2) *Design of the Correction Gain P_{cor}* : From the above analyses, it can be seen that the polarity correction function works only when the system converges to incorrect convergence points $((2k+1)\pi, 0)$, in which case the open-loop transfer function of GPLL is (assumed $\Psi_f/\hat{\Psi}_f = 1$)

$$G_{O-NEW}(s) = \frac{(fk_p + \omega_c)s^2 + f(k_i + k_p\omega_c)s + fk_i\omega_c}{s^2(s + \omega_c)} \quad (25)$$

where f is the polarity correction function $f_{\alpha\beta(dq)}$. At this time, we get the cross-over frequency of the GPLL ω_p by

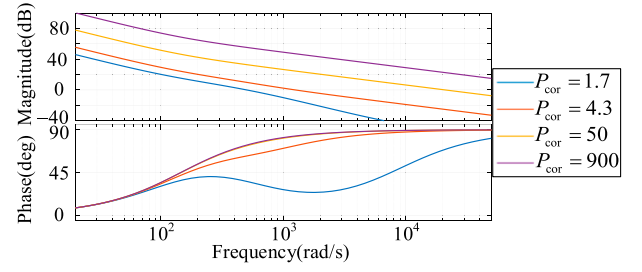


Fig. 8. Bode diagrams of the GPLL at different P_{cor} .

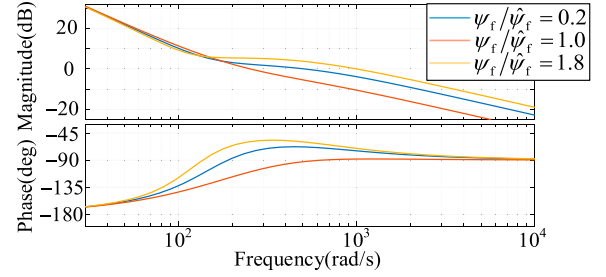


Fig. 9. Bode diagrams of the GPLL with different $\Psi_f/\hat{\Psi}_f$.

$$|G_{O-NEW}(j\omega_p)| = 1. \quad (26)$$

Since the GPLL bandwidth needs to be four to ten times greater than ω_{spd} [7], and the correction process requires faster response ability than the noncorrection process, we take $\omega_p = 10 \times \omega_{spd} = 200 \times 2\pi$ rad/s. By replacing ω_p and $\omega_{PLL} = 40 \times 2\pi$ rad/s into (26), we can get that $f_{\alpha\beta(dq)}$ is about 7.6, i.e., $P_{cor} = 4.3$, which can improve the dynamic performance of the correction process and meanwhile not sacrifice the steady-state performance.

The Bode diagrams for the GPLL systems with different correction gains P_{cor} under $\omega_{PLL} = 40 \times 2\pi$ rad/s and $\omega_c = 80 \times 2\pi$ rad/s according to (26) are shown in Fig. 8. Compared with Fig. 7, the dynamic performance is significantly improved with the increase of the P_{cor} . However, the corresponding high-frequency noise suppression ability is decreased. In addition, it can be seen that when $P_{cor} > 50$, the phase-frequency diagrams are basically coincident. Therefore, when the P_{cor} reaches a certain level, the steady-state performance of the system is not affected.

To sum up, because the correction gain does not affect the system performance at the correct convergence domain, it is suggested to choose a slightly larger gain in the incorrect convergence domain, i.e., between $2 < |f_{\alpha\beta(dq)}| < 10$, which means $1.5 < P_{cor} < 5.5$, to accelerate the system back to the correct convergence domain.

3) *The Effect of $\Psi_f/\hat{\Psi}_f$ on the Performance of the GPLL*: Actually, $\Psi_f/\hat{\Psi}_f$ would not be equal to one all the time, considering the temperature variation and magnetic field saturation of permanent magnets [19]. Hence, we need to consider the influence of $\Psi_f/\hat{\Psi}_f \neq 1$ on the tracking performance of the GPLL. Fig. 9 shows the Bode diagram of the GPLL with different $\Psi_f/\hat{\Psi}_f$ values at $\omega_{PLL} = 40 \times 2\pi$ rad/s and $\omega_c = 80 \times 2\pi$ rad/s. As shown in Fig. 9, the larger $\Psi_f/\hat{\Psi}_f$ is, the better the dynamic performance of the system is, while the steady-state



Fig. 10. Experimental platform.

TABLE I
MAIN PARAMETERS OF THE TEST PMSM

Parameters	Values	Parameters	Values
Rated voltage	48 V	Stator resistance	0.15 Ω
Rated current	10 A	L_d	0.193 mH
Rated speed	3000 r/min	L_q	0.193 mH
Flux linkage	0.0116 Wb	Pole pairs	5

TABLE II
SYSTEM CONTROL PARAMETERS

Parameters	Values	Parameters	Values
LO bandwidth	$800 \times 2\pi$ rad/s	GPLL bandwidth	$80 \times 2\pi$ rad/s
SMO gain	1.0	Speed-loop bandwidth	$20 \times 2\pi$ rad/s
Traditional PLL bandwidth	$80 \times 2\pi$ rad/s	Feed-forward ω_c	$80 \times 2\pi$ rad/s
S-PLL bandwidth	$80 \times 2\pi$ rad/s	Correction gain P_{cor}	4.3

performance decreases. Moreover, the magnitude plots in Figs. 7 and 9 all coincide in the low-frequency range, which means the robustness of the system is not affected even at different ω_c and the error existence of $\hat{\Psi}_f$.

IV. ANALYSIS OF EXPERIMENTAL RESULTS

The effectiveness of the GPLL is verified by comparing the GPLL with some other methods in various complex conditions on a PMSM drive platform shown in Fig. 10. This platform is composed of a 0.4 kW PMSM and a magnetic powder brake which provides resistance loads, etc.

The parameters of the test PMSM are given in Table I. The main control chip is a TMS320F28335 and the PWM frequency is 20 kHz. The control parameters are shown in Table II. Under the same parameters, the traditional PLL, S-PLL, and the GPLL were compared both adopting the SMO on $\alpha\beta$ -axis and the LO on dq -axis, respectively. As a comparison method, AFO method adopts the method in reference [8].

A. Comparison of Speed Reversal Experiments at Medium and Low Speeds

The experiments are all carried out under a rated load in this section, and the experimental condition is that the speed reference ramp decreases from 600 to -600 r/min.

1) *Speed Reversal Experiment of the Traditional PLL:* Fig. 11 shows the speed reversal experiments of the traditional

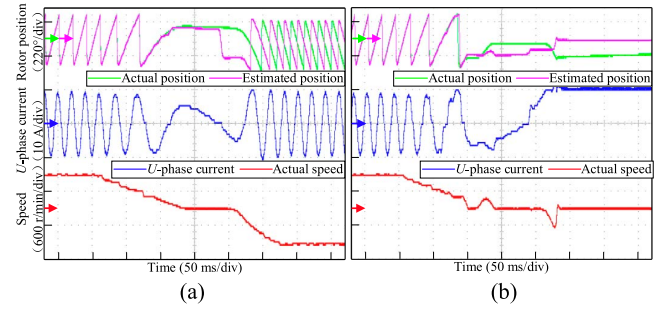


Fig. 11. Speed reversal test of the traditional PLL. (a) Sensor control. (b) Sensorless control.

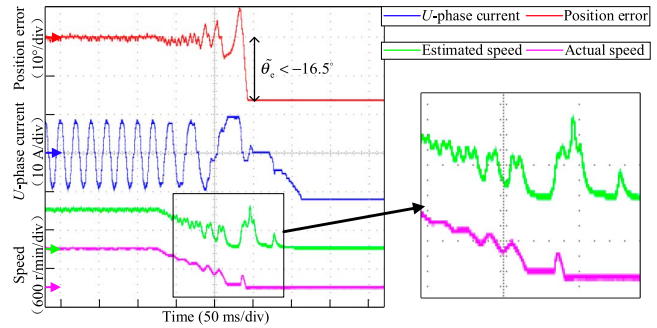


Fig. 12. Failed speed reversal test of the S-PLL.

PLL. It is seen from Fig. 11(a), in sensor control, the position tracking error of the traditional PLL is close to 0° when the motor is running in a forward direction. When the motor direction changes from positive to negative, the position tracking error stays at 180° . Meanwhile, from Fig. 11(b), when the motor spins in a negative direction in sensorless control, the motor has eventually stopped running due to the incorrect convergence of the traditional PLL.

2) *Speed Reversal Experiment of the S-PLL and the AFO:* In contrast, the S-PLL method is tested under the same experimental conditions. After repeating the same experiment many times, we found that the S-PLL has only a 20% success rate at speed reversal test. It is affected by the steady-state error in estimating speed at very low speed, resulting in the speed symbol bouncing back and forth between positive and negative, which induces a very high probability of the S-PLL getting out of control near zero speed. As shown in Fig. 12, when the actual speed tries to rotate from zero speed to negative speed, the S-PLL incorrectly estimates speed as a positive value, resulting in a “positive-feedback system,” which makes the system becomes completely out of control. (The “Position error” in Fig. 12 is limited by the maximum output of the chip DAC, and its highest value is 16.5° .)

Fig. 13(a) shows the experiment when the S-PLL successfully crosses zero speed. It can be found that the S-PLL stays at standstill for a long time during the reversal process. Because it is not only affected by the weaker magnitude of the BEMF, but also affected by the oscillation of the speed symbol.

For comparison, the performance of the AFO was verified under the same experimental conditions, as shown in Fig. 13(b). It can be seen that the rotor stays at zero speed for a long time,

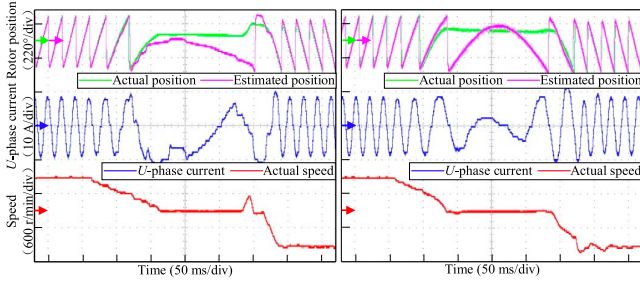


Fig. 13. Speed reversal test of (a) S-PLL and (b) AFO.

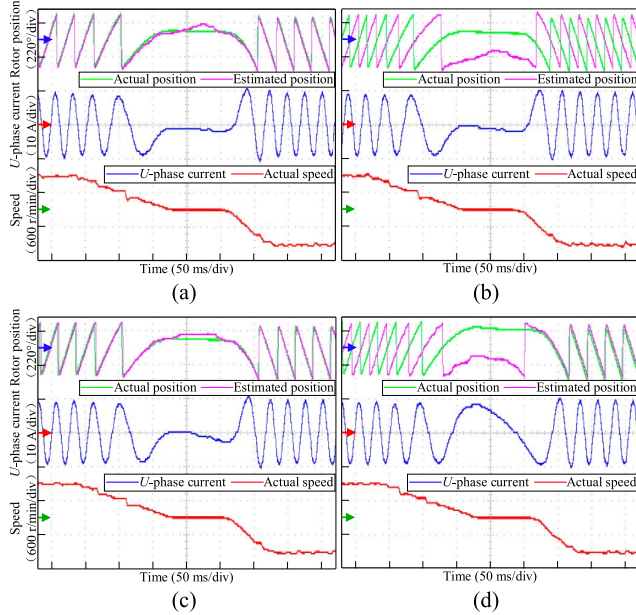


Fig. 14. Speed reversal test of the GPLL without a polarity correction in sensor control. (a) and (b) Using the LO. (c) and (d) Using the SMO.

the dynamic ability is obviously weak, but its waveform is relatively gentle.

3) *Speed Reversal Experiments of the GPLL*: Fig. 14 shows the speed reversal experiments of the proposed GPLL without a polarity correction in sensor control. Fig. 14(a) and 14(b), 14(c) and 14(d) all repeated the experiment twice at the same condition, respectively. As can be seen from Fig. 14, no matter on dq - or $\alpha\beta$ -axis, and in positive or negative rotation, the GPLL without a polarity correction may converge to the correct position or the incorrect position with 180° error.

Fig. 15 shows the speed reversal experiments of the GPLL with a polarity correction in sensorless control. In the process of speed reversal, it can be seen from (a)–(c) adopting the LO, when $P_{cor} = 1.0$, the motor rotor has a long-time lockup and has poor dynamic response ability. When $P_{cor} = 4.3$, the speed becomes smooth and stable in transition. When $P_{cor} = 10.5$, the estimated position is two electrical angle cycles ahead of the actual position, causing a violent oscillation of the speed. Compared with the experiments of (d)–(f), it is observed that the motor all achieves smooth and stable operations based on the SMO in regardless of the value of P_{cor} .

In addition, after repeating the above experiments for many times, the conclusion is drawn that the GPLL with a polarity

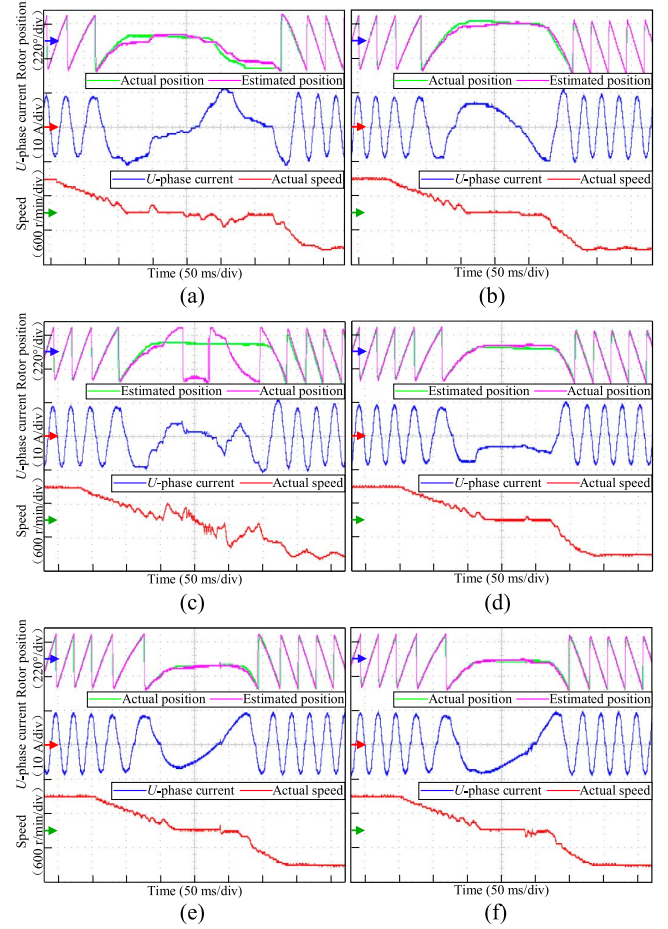


Fig. 15. Speed reversal test of the GPLL with a polarity correction in sensorless control. (a), (b), and (c) Using the LO at $P_{cor}=1.0, 4.3$, and 10.5 , respectively. (d), (e), and (f) Using the SMO at $P_{cor}=1.0, 4.3$, and 10.5 , respectively.

correction can realize speed reversal each time. Meanwhile, compared with Figs. 13(a) and 13(b), the transition process of the experiment in Fig. 15(b) is significantly smoother, with almost no position error. This is all because the GPLL is completely unaffected by the speed symbol compared to the S-PLL. Meanwhile, the dynamic performance of BEMF-POs is better than that of AFO.

On the whole, the performance of the GPLL is affected by the correction gain and the performance of the BEMF-POs. If the correction gain is too small, the position tracking convergence is slow when the speed crosses zero, resulting in the speed oscillation or even lockup near zero speed. If the correction gain is too large, it will lead to a large position tracking error when the motor speed crosses zero, resulting in the obvious oscillation of the speed. Therefore, selecting an appropriate correction gain can not only bring no oscillation and no lockup during the motor speed crossing zero but also ensure the position tracking error is small. In addition, because the nonlinear observer, such as the SMO in this case, has strong robustness, the change of the correction gain in a certain range will not affect the crossing-zero performance greatly.

B. Performance Comparison Experiment at Very Low Speed

To fully demonstrate the performance of the GPLL, speed reversal experiments without load at very low speeds are executed, where the effect of voltage nonlinearities is nearly equivalent to the magnitude of the BEMF, which is the most difficult condition for observers. In this section, only the LO is chosen as an example to demonstrate the performance of the GPLL due to the article's length.

At the beginning, the minimum speed of the LO based on the GPLL method can only reach 300 r/min. And after several experiments, it is found that only 50% of the probability can successfully achieve speed reversal at 300 r/min. Furthermore, we also found that the higher the speed or the larger the load, the higher the probability of speed reversal. Under 600 r/min with rated load condition, each time can successfully achieve speed reversal. This is because the lower the speed or lighter load, the more influence of the inverter nonlinearities and stator resistance error on the BEMF estimation.

In contrast, the speed reversal experiments on the S-PLL from 300 to -300 r/min without load are conducted. However, after many experiments, it was unable to achieve the speed reversal successfully. The reason for this is the same as that in Fig. 12. The only difference is that the experiment in Fig. 12 is carried out under rated load, the current amplitude is large, and the speed is fast, so the influence of voltage nonlinearities is smaller, and the success rate is higher.

To further address the problem that speed reversal probability is not 100% success rate and to broaden the range of the observer at lower speeds, two solutions are adopted: 1) when the speed is lower than 600 r/min, a current with amplitude of 1 A (10% of the rated current) is injected into the estimated d -axis. This method can greatly improve the signal-to-noise ratio of the BEMF or flux estimation and relieve the influence of stator resistance error on the BEMF or flux estimation so as to improve the antinoise ability and stability of the observer during the speed reversal [10], [20]. 2) Add a dead-time compensation to directly eliminate nonlinearities of the inverter. This is considered as the best method to improve the performance of the observer at extremely low speed. The inverter nonlinearities compensation method in [21] is adopted in this article.

Fortunately, adopting the above two tricks, it is found that the GPLL-based LO could achieve a minimum speed reaches 30 r/min, and still achieve a 100% successful speed reversal at this speed, as shown in Fig. 16(a). We can see that it has a steady-state error of -6° and 5° because the estimation of the BEMF is greatly degraded at very low speed, which is unavoidable. In addition, the current has no obvious impact, but the speed has slight overshoot. Because the speed is too low, the speed estimation is not very accurate during the reversal process.

In comparison, the performance of the S-PLL under the same experimental conditions is carried out, but the minimum speed of the S-PLL is only about 90 r/min even after injecting the same d -axis current and adding dead-time compensation, and the speed reversal success rate is only about 30% at 90 r/min. This is because the speed estimation will become negative when

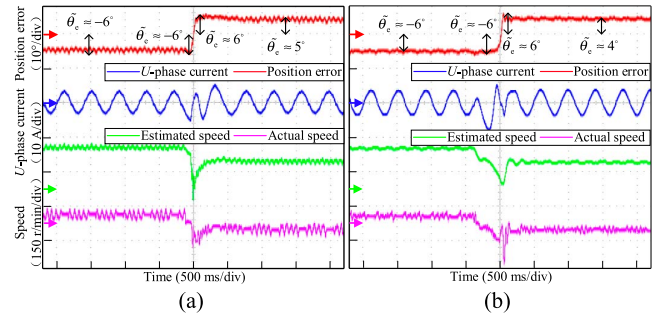


Fig. 16. Speed reversal test at 30 r/min without load based on (a) GPLL and (b) AFO.

there is a slight steady-state error at very low speed, which forms a “positive-feedback system,” and leads to a divergence. After all, the above two tricks can only improve position and speed estimation accuracy to some extent, but cannot totally solve the influence of the speed symbol.

Finally, speed reversal experiments based on the AFO with the same d -axis current injection and dead-time compensation were tested at 30 r/min without load, as shown in Fig. 16(b). Compared with Fig. 16(a), the experimental results based on the AFO have some differences: the fluctuation of the actual speed in the steady state is significantly smaller than that of the LO. Meanwhile, the steady-state position errors are -6° and 4° , which are slightly smaller than those of the LO. However, the speed fluctuation of the AFO during reversal is larger than that of the LO.

To sum up, after completely eliminating the influence of the speed symbol, the BEMF-POs based on the GPLL can also achieve a relatively ideal and similar performance with those of the AFO under very low-speed and speed reversal conditions by adding dead-time compensation and injection d -axis current.

C. Dynamic Performance Experiment

1) *Influence Analysis of the Cutoff Frequency of the LPF and Flux Linkage Error on Sensorless Control Performance:* In the previous section, Fig. 15(b) and 15(e) shows the experiments under the conditions of $P_{cor} = 4.3$, $\omega_c = 80 \times 2\pi$ rad/s, and $\Psi_f/\hat{\Psi}_f = 1$. To evaluate the influence of different ω_c and flux linkage errors on the tracking performance of the GPLL, the comparative experiments were conducted as follows.

Fig. 17 shows the experiments of the GPLL in sensorless control with the $P_{cor} = 4.3$ and $\Psi_f/\hat{\Psi}_f = 1$, when the speed linearly decreases from 600 to -600 r/min at the rated load. During the speed reversal, it can be seen from (a) and (b) that when $\omega_c = 20 \times 2\pi$ rad/s, the dynamic response is slow, and the motor locks up for a long time during the crossing-zero process. When $\omega_c = 140 \times 2\pi$ rad/s, although the reversal process is faster, the speed in the steady-state after the reversal has an oscillation. Similarly, it can be seen from (c) and (d) that when $\omega_c = 20 \times 2\pi$ rad/s, the dynamic response of the system slows down significantly. When $\omega_c = 140 \times 2\pi$ rad/s, the dynamic response of the system is fast, but the speed oscillates. However, when $\omega_c = 80 \times 2\pi$ rad/s, as shown in Fig. 15(b) and 15(e), the system has smooth and rapid transition, and its dynamic and steady performance is good. In summary, when the ω_c is low, the dynamic

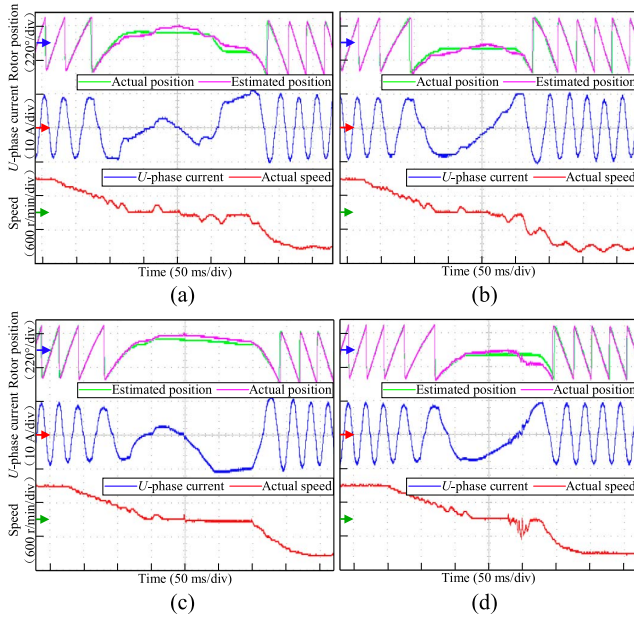


Fig. 17. Influence of the ω_c on the control performance. (a) LO with $\omega_c = 20 \times 2\pi$ rad/s. (b) LO with $\omega_c = 140 \times 2\pi$ rad/s. (c) SMO with $\omega_c = 20 \times 2\pi$ rad/s. (d) SMO with $\omega_c = 140 \times 2\pi$ rad/s.

response capability of the system is poor. As the ω_c increases, although the dynamic response becomes faster, the steady-state control performance of speed is weakened.

Fig. 18 shows the experiments of the GPLL in sensorless control with the $P_{cor} = 4.3$ and $\omega_c = 80 \times 2\pi$ rad/s when the speed ramp decreases from 600 to -600 r/min at the rated load. It can be seen from Fig. 18(a) and 18(c) that when the $\Psi_f/\hat{\Psi}_f$ of the GPLL is less than 1, the dynamic response ability of the GPLL slows down obviously. In (b) and (d), when the $\Psi_f/\hat{\Psi}_f$ is greater than 1, the speed oscillated significantly after the reversal, so its steady-state performance decreased significantly. In summary, the flux linkage error will also affect the system's performance. The larger the $\Psi_f/\hat{\Psi}_f$ value, the stronger the dynamic response capability of the system, but the steady-state performance is significantly reduced.

2) *Feedforward Compensation Performance Evaluation:* Since the transfer functions of the GPLL on both dq - and $\alpha\beta$ -axes are the same, and due to the article length limitation, only the performance of the GPLL on dq -axis is analyzed.

The experimental condition in Fig. 19 is: the rated load is suddenly added at 600 r/min, then the speed is linearly accelerated to 3000 r/min, and the rated load is suddenly removed, and finally, the speed linearly decelerated to 600 r/min. As can be seen from (a), the maximum position tracking error of the traditional PLL reaches 12° and -12° during speed acceleration and deceleration, respectively. In contrast, in terms of dynamic tracking performance, the GPLL has almost no position tracking error during speed transient processes. In addition, the steady-state error curve of the traditional PLL is obviously thickened when the rated load is suddenly increased, while that of the GPLL is unchanged. When the load is suddenly removed, the position tracking error of the traditional PLL reaches 50° and causes the speed oscillation, while the error of the GPLL is only

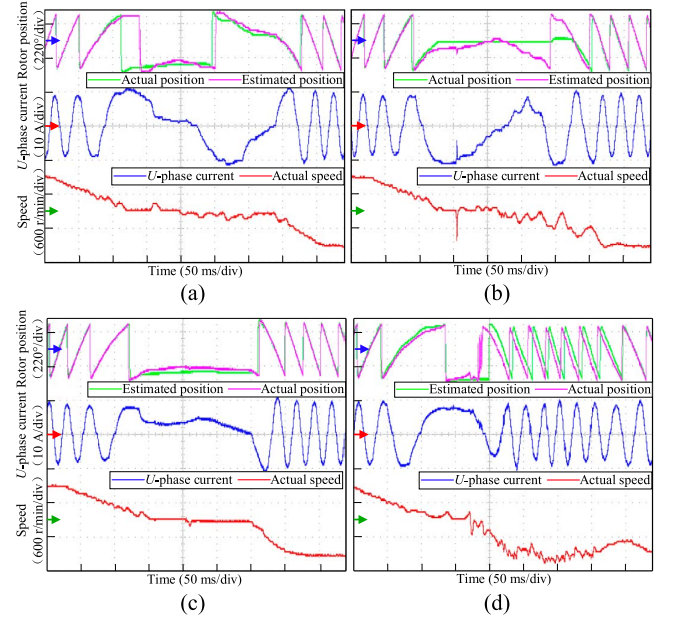


Fig. 18. Influence of flux linkage error on the sensorless control performance. (a) LO with $\Psi_f/\hat{\Psi}_f = 0.6$. (b) LO with $\Psi_f/\hat{\Psi}_f = 1.4$. (c) SMO with $\Psi_f/\hat{\Psi}_f = 0.6$. (d) SMO with $\Psi_f/\hat{\Psi}_f = 1.4$.

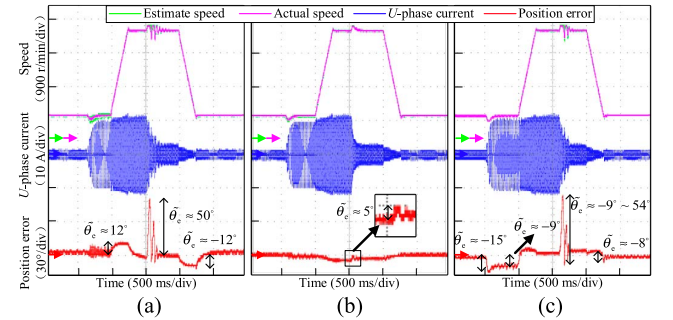


Fig. 19. Comparison of the dynamic performance between: (a) traditional PLL; (b) proposed GPLL; and (c) AFO.

5° and without speed oscillation. In fact, when the position tracking error reaches a large amount in a short period of time, the convergence domain of the PLL system is likely to be incorrect. So, the polarity correction function may play an important role, and it is why the dynamic performance of the GPLL is so good.

In comparison, a similar experiment under the same conditions using the AFO is conducted, and the maximum position tracking error reaches -10° and -8° during speed acceleration and deceleration, respectively, as shown in Fig. 19(c), which is slightly better than the traditional PLL. However, the position tracking error of the AFO reaches 15° when the rated load is suddenly increased and 54° when the rated load is suddenly removed, which is worse than that of the GPLL.

Finally, to show that the GPLL does not have a great impact on the steady-state performance of the system, the position tracking errors of the three methods under rated load and rated speed are shown in Fig. 20. Compared with Fig. 20(a) and 20(b), the steady-state error of the S-PLL is only -3.5° – 0° , while that of the GPLL reaches -4° – 0.5° . It is because the GPLL combines

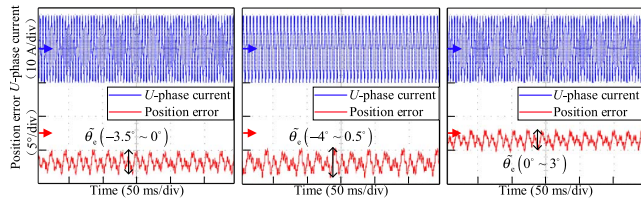


Fig. 20. Comparison of the steady-state performance between: (a) traditional PLL; (b) proposed GPLL; and (c) AFO.

the speed feedforward compensation. However, this small difference will not affect the steady-state performance. As can be seen from Fig. 20(c), the steady-state error of the AFO is only $0^\circ\sim 3^\circ$, which is slightly better than the other two methods. This is because the AFO itself has a better steady-state estimation accuracy, and its algorithm implementation is a second-order filter [9].

V. CONCLUSION

To overcome the two shortcomings of the traditional PLL in the sensorless control of PMSMs: incapability in speed reversal and an unsatisfactory position tracking performance between the dynamic and steady-state states, a novel GPLL, which is applicable to both stationary $\alpha\beta$ -axis and rotary dq -axis, is proposed and the design process and its criteria are also presented. It has the following advantages.

- 1) The input error signal of the GPLL is constructed to eliminate the influence of the sign of the speed variable so that the GPLL can accurately track the actual position in both positive and negative rotation conditions.
- 2) Using the relationship between the estimated BEMF and the estimated speed, a polarity correction function is designed to solve the problem of multiple convergence points in the GPLL. It also avoids the effect of the speed symbol and makes the GPLL to achieve lower operation speed and more reliable speed reversal with 100% success rate with the help of methods such as dead-time compensation and d -axis current injection. Moreover, the larger polarity correction gain, the stronger the dynamic performance in the position tracking; however, the speed oscillates larger in steady state.
- 3) The estimated speed feedforward compensation is added to the GPLL so that the dynamic position tracking error can be reduced, and the dynamic performance can be enhanced under the conditions of acceleration and deceleration, loading and unloading. In the feedforward path, the cutoff frequency of the LPF and flux linkage error will affect the dynamic performance of the position tracking.

Finally, through a series of comparative experiments, it is verified that the GPLL with the polarity correction function can achieve 100% speed reversal at medium and low speeds, and its flatness and success rate are stronger than those of the S-PLL. Furthermore, the dynamic performance of GPLL with feedforward compensation is significantly stronger than that of the conventional PLL-based methods; meanwhile, the steady-state accuracy is not affected. The lowest speed of the LO based on the GPLL with the d -axis current and dead-time compensation

can reach 1% of the rated speed at no load, which is as good as that of the AFO method.

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